

Midterm practice problems

This collection provides additional practice with the topics in PS1 and PS2. It is not intended to match the length or structure of the midterm, and it is not a timed practice exam. Work through the problems at your own pace, using the worked solutions to examine the reasoning behind each step after you have attempted a solution.

For some questions, several approaches may be appropriate. Explain how the objective, available information, and constraints support your choice.

The written problems are followed by an optional programming exercise. This collection is for preparation, not an additional graded submission. You may enter answers and plots in the `studentanswer` environments of `midterm-practice/handout.tex`, or write on the PDF. Solutions are distributed separately.

AI Policy. You may use any AI tools you like to assist you with your assignments, and you should, as that is likely how you will work for the rest of your career. That being said, the written portions are practice for the in-class midterm and exam, which are closed book and on paper. You should strive to answer these yourself and use AI as a tutor and not an oracle.

Please include an AI Tool Disclosure Below

The formula sheet is included for reference and will be included in the exam.

Formula sheet

Search and planning

$$f(n) = g(n) + h(n), \quad h(n) \leq h^*(n), \quad h(n) \leq c(n, n') + h(n'), \quad h(G) = 0$$

Break A* ties by lower h , then row-major order.

MDPs and RL

$$(\mathcal{S}, \mathcal{A}, T, R, \gamma), \quad T(s' | s, a) = P(s' | s, a), \quad \sum_{s'} T(s' | s, a) = 1, \quad 0 \leq \gamma < 1$$

$$Q^*(s, a) = \sum_{s'} T(s' | s, a) [R(s, a, s') + \gamma \max_{a'} Q^*(s', a')]$$

$$V^*(s) = \max_a Q^*(s, a), \quad \pi^*(s) \in \arg \max_a Q^*(s, a)$$

$$V_{k+1}(s) = \max_a \sum_{s'} T(s' | s, a) [R(s, a, s') + \gamma V_k(s')]$$

$$V^{\pi_k}(s) = \sum_{s'} T(s' | s, \pi_k(s)) [R(s, \pi_k(s), s') + \gamma V^{\pi_k}(s')]$$

$$\pi_{k+1}(s) \in \arg \max_a \sum_{s'} T(s' | s, a) [R(s, a, s') + \gamma V^{\pi_k}(s')]$$

$$V(s) \leftarrow V(s) + \alpha [r + \gamma V(s') - V(s)]$$

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Optimization

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k), \quad f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2} f''(x_0)(x - x_0)^2$$

Transforms

$${}^a\mathbf{T}_b = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} p_a \\ 1 \end{bmatrix} = {}^a\mathbf{T}_b \begin{bmatrix} p_b \\ 1 \end{bmatrix}, \quad {}^a\mathbf{T}_c = {}^a\mathbf{T}_b {}^b\mathbf{T}_c, \quad R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Calculus

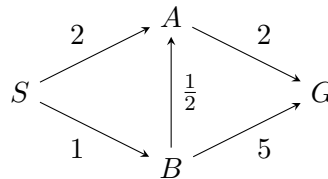
$$(x^n)' = nx^{n-1}, \quad (\sin x)' = \cos x, \quad (\cos x)' = -\sin x$$

$$\sin 0 = 0, \quad \cos 0 = 1, \quad \sin(\pi/2) = 1, \quad \cos(\pi/2) = 0$$

Part 1: Written practice

Question 1: Changing a heuristic

The directed graph has edges $S \rightarrow A$ of cost 2, $S \rightarrow B$ of cost 1, $B \rightarrow A$ of cost $\frac{1}{2}$, $A \rightarrow G$ of cost 2, and $B \rightarrow G$ of cost 5. There are no other edges. The heuristic is $h(S) = 3$, $h(A) = 2$, $h(B) = 1$, $h(G) = 0$.



- (a) Find the optimal cost from each node to G . Is the heuristic admissible? Justify by comparing the values.

Answer / work

- (b) Is it consistent? Identify a specific edge inequality that establishes your answer.

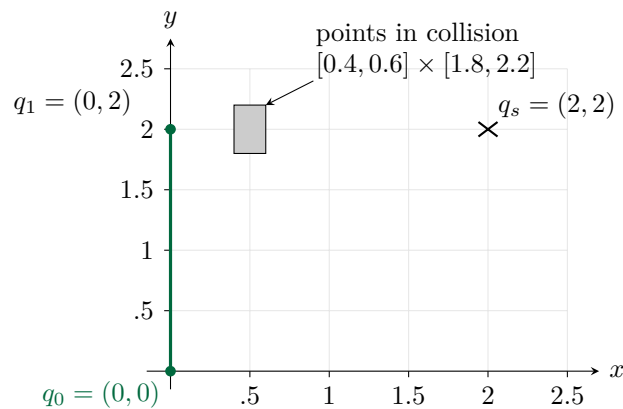
Answer / work

- (c) Change only $h(B)$ to $\frac{5}{2}$. Is the resulting heuristic both admissible and consistent? Check all edges incident to B and explain why the remaining edges do not create a new violation.

Answer / work

Question 2: Validating an RRT extension

A two-dimensional configuration-space RRT currently contains nodes $q_0 = (0, 0)$ and $q_1 = (0, 2)$. The next sample is $q_s = (2, 2)$ and the maximum extension length is 1, using Euclidean distance and straight-line steering. The set of points in collision is the rectangle $[0.4, 0.6] \times [1.8, 2.2]$. Touching its boundary counts as collision. There are no other obstacles. The drawing shows the existing tree and the sample, but not the proposed extension.



- (a) Identify the nearest node and compute the proposed new node for maximum extension length 1. Is that endpoint itself in collision? Show your work. Check the connecting segment separately in part (b).

Answer / work

- (b) For the same length-1 extension, determine whether the entire segment from the nearest node to the proposed endpoint is collision-free. Should RRT accept the extension? Explain why an endpoint check alone is insufficient, and state what the tree contains after this iteration.

Answer / work

- (c) Starting independently from the original two-node tree in each case, compare maximum extension lengths $\frac{1}{4}$, 1, and 2 for the same sample. For each length, give the proposed endpoint, state whether the endpoint is in collision, and determine whether the entire connecting segment is collision-free. Which extensions would RRT accept? Your length-1 result should agree with parts (a) and (b). Does this comparison establish that a smaller or larger maximum extension is preferable for the planning problem as a whole? Explain.

Answer / work

Question 3: Learning from recorded episodes

A fixed policy generates the following three episodes. The state Z is terminal, arrows show observed rewards, and $\gamma = \frac{1}{2}$. The episodes are listed in the order they were recorded.

Episode	Observed transitions
1	$A \xrightarrow{2} B \xrightarrow{4} Z$
2	$A \xrightarrow{0} B \xrightarrow{2} Z$
3	$A \xrightarrow{1} B \xrightarrow{6} Z$

- (a) Compute the discounted return from A and from B in each episode. Then compute the direct-evaluation estimate of each state's value.

Answer / work

- (b) Start $V(A) = V(B) = 0$ and process the transitions in their recorded order with TD learning rate $\alpha = \frac{1}{2}$. Keep the estimates between episodes and hold $V(Z) = 0$. For each of the six transitions, show the sample $r + \gamma V(s')$ and the updated value of the state being left. Give the final table.

Answer / work

- (c) Explain why these estimates can differ without either calculation being incorrect. Identify the continuation information each uses.

Answer / work

Question 4: Evaluation is not improvement

An MDP has states A, B, Z , with terminal value $V(Z) = 0$ and $\gamma = \frac{1}{2}$. At A , action *stop* gives reward 1 and goes to Z . Action *continue* gives reward 0 and goes to B . At B , the only action gives reward 4 and goes to Z . All transitions are deterministic.

- (a) Evaluate the policy that stops at A . Give $V^\pi(A)$ and $V^\pi(B)$.

Answer / work

- (b) Perform one policy-improvement step at A , showing the value assigned to each candidate action. Does the policy change?

Answer / work

- (c) Independently initialize $V_0(A) = V_0(B) = 0$ and perform two synchronous VI sweeps. Give V_1 and V_2 . Explain why the first sweep is not the same operation as exact policy evaluation followed by improvement.

Answer / work

Question 5: Descending on an approximation

Let $f(x) = x^4 - 2x^2 + x + 1$. Consider its quadratic Taylor approximation $m(x)$ about $x_0 = 1$.

- (a) Evaluate $f(1)$, $f'(1)$, and $f''(1)$ and construct $m(x)$.

Answer / work

- (b) Starting at $x = 1$, take one gradient step on m with $\alpha = \frac{1}{8}$. Show that the result minimizes m , but is not a stationary point of f . What does this show about interpreting a step computed from a local model?

Answer / work

Question 6: Planning to make coffee

A six-joint arm must move a mug around a cluttered counter. Joint limits and a geometric collision checker are available. A new arrangement of objects is used for each task. Consider RRT, PRM, A*, trajectory optimization, VI, and Q-learning. More than one choice can be defended. State the assumptions supporting your answer.

- (a) Propose an approach for moving the mug between two specified arm configurations in one arrangement. Justify your choice and compare it with a plausible alternative. State any additional assumptions your approach requires.

Answer / work

- (b) The resulting path avoids the objects, but the planned motion changes direction abruptly and is unsuitable for carrying a full mug. How would you revise your approach? Explain what you would change and what evidence would be needed before accepting the revised motion.

Answer / work

- (c) The robot must choose whether to fill the mug before or after moving it. A small discrete model gives transition probabilities and rewards, including the consequences of spills. Propose a method for making this decision, justify it relative to an alternative, and explain how it would fit into your overall approach to the task.

Answer / work

Question 7: Same tool position, different configurations

A planar two-link arm has unit-length links, a fixed base at $(0,0)$, and two revolute joints. Its configuration is $q = (\theta_1, \theta_2)$, where the second angle is relative to the first link. Configuration $q_A = (0, \pi/2)$ has elbow $(1,0)$ and tool position $(1,1)$. Configuration $q_B = (\pi/2, -\pi/2)$ has elbow $(0,1)$ and the same tool position $(1,1)$. Treat links as zero-width segments. The only obstacle is a disk of radius 0.1 centered at $(1,0)$. Draw the two arm configurations as needed to support your reasoning.

- (a) What is the dimension of configuration space? Explain why the tool position alone does not specify the configuration in this example.

Answer / work

- (b) Which of q_A, q_B is collision-free? Explain why checking only the tool point would give an incorrect safety conclusion.

Answer / work

Question 8: Interpreting a learning experiment

A robot follows one fixed policy and records complete episodes. Its engineer first wants to estimate that policy’s expected return, not find a different policy. Later, the engineer trains a Q-learner using an exploratory behavior policy and reports the reward collected during training.

- (a) Compare direct evaluation and TD V-learning for the initial objective. State what each uses as a target and one reason to prefer either method in a specified data-collection setting. Is Q-learning with a maximizing target evaluating the same fixed policy?

Answer / work

- (b) Does high or low reward during exploratory training by itself establish how well the final greedy policy performs? Describe a separate evaluation that would support that conclusion without changing the learned table, and identify one limitation of the resulting evidence.

Answer / work

Question 9: What a value table cannot identify

An agent is given a value table but no transition or reward model. Each MDP below has one nonterminal state s , two actions a, b , and a terminal state. Both actions terminate immediately. Rewards are deterministic.

- (a) Construct two such MDPs with the same $V^\pi(s)$ for $\pi(s) = a$ and identical observed episodes under π , but different uniquely optimal actions. Explain why arbitrarily many episodes under that policy cannot distinguish the MDPs.

Answer / work

- (b) Could exact V^* alone, without a model, always identify an optimal action? Give another pair of MDPs to establish your answer. Explain why being given exact Q^* changes the action-selection problem.

Answer / work

Question 10: Special cases of search

- (a) Run A^* with $h \equiv 0$. Which classic algorithm are you now running?
- (b) Under what *two* conditions on h and the edge costs does A^* reduce to breadth-first search?

Answer / work

Question 11: Value iteration on a chain

Three states in a row: $s_1 \leftrightarrow s_2 \rightarrow s_3$. From s_1 you may move right to s_2 or stay. From s_2 you may move right into s_3 or left to s_1 . Entering s_3 (the goal, absorbing, $V(s_3) \equiv 0$) earns reward +8. Every other transition earns 0. Discount $\gamma = 0.5$. Starting from $V_0 = 0$ everywhere, run **two** synchronous value-iteration sweeps and give V_1 and V_2 for s_1 and s_2 .

Answer / work

Question 12: Scaling a grid heuristic

On a 4-connected unit-cost grid, consider $h_1 =$ Manhattan distance to the goal and $h_2 = 2 \times$ Manhattan distance.

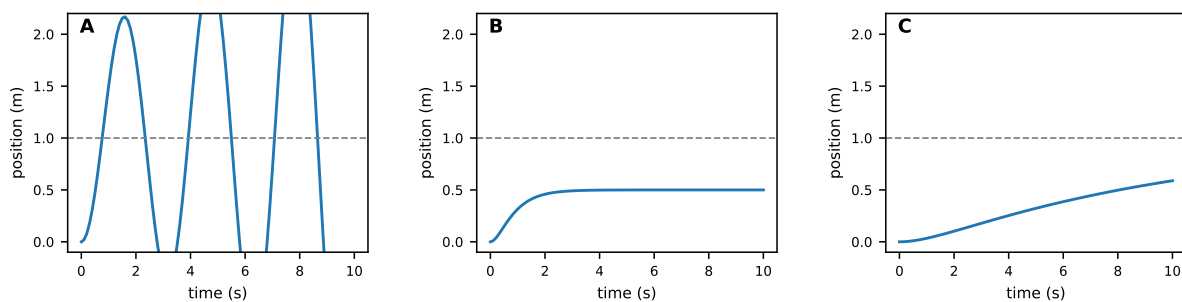
- Which are admissible? Justify using the cell adjacent to the goal.
- One sentence: what can A^* return if you run it with h_2 ?

Answer / work

Question 13: Diagnosing controller behavior

A point-mass robot is commanded to move to position $p = 1$ m. The plots below show its position over time for three different controller configurations. For each plot, compare the observed behavior with the following candidate causes:

- (1) position feedback only, no damping (no velocity or derivative term);
- (2) a constant disturbance (the test strip is on a slope) and no integral action;
- (3) an excessively large effort weight $\mathbf{R}$, giving weak control;
- (4) an excessively small effort weight $\mathbf{R}$, causing actuator saturation;
- (5) the reference is not reachable under the control limits.



- (a) For plot A, identify the most likely cause from the list, justify the diagnosis, and propose a correction.

Answer / work

- (b) For plot B, identify the most likely cause from the list, justify the diagnosis, and propose a correction.

Answer / work

Answer / work

Question 14: Policy evaluation and optimal control

Consider finite discounted Markov decision processes with $0 < \gamma < 1$. Here, V-learning means tabular TD evaluation of a fixed policy π . Value iteration (VI) and policy iteration (PI) have access to the exact transition and reward model; V-learning and Q-learning use sampled transitions. All values are tabular, and terminal states have value zero. Distinguish the information needed to learn a value function from the information needed to select an action once that value function is known. In parts (b) and (c), all rewards are deterministic and each action leads directly to the terminal state. You may choose any real-valued rewards.

- (a) Compare VI, PI, V-learning, and Q-learning in a table or four short paragraphs. For each method, identify the stored value function and any explicit policy, the information used in an update, and whether the objective is fixed-policy evaluation or optimal control. Explain whether the successor value follows the current policy or includes a maximization over actions. Then describe the two stages of PI and explain why completing policy evaluation alone does not complete policy iteration.

Answer / work

- (b) A student claims: “Once V-learning has recovered V^π exactly, we can identify an optimal action in every state without a transition or reward model.” Disprove this claim using two MDPs with one nonterminal state s , two actions a, b , and one terminal state. Both actions terminate immediately, and $\pi(s) = a$. Your MDPs must have the same $V^\pi(s)$ but different uniquely optimal actions. Explain why observing only this policy does not allow a learner to distinguish the two MDPs. State the reward of each action in each MDP, compute $V^\pi(s)$, and identify the unique optimal action in each. Finally, explain why additional episodes under the same fixed policy do not resolve the ambiguity. Your argument should concern missing information, not estimation error or slow convergence.

Answer / work

- (c) Suppose instead that the learner knows V^* exactly but still has no model. Does this resolve the problem? Construct two one-step MDPs with the same $V^*(s)$ but different uniquely optimal actions. Then explain what additional information lets VI and PI extract a greedy policy, and why Q^* itself is sufficient without a model. Give both reward tables, compute the optimal values, and explicitly identify the optimal actions. Write the model-based one-step lookahead expression used for policy extraction and the corresponding expression when Q^* is available. Explain why being given an exact function is different from having enough experience to learn it accurately.

Answer / work

Question 15: Inflated heuristics

A* normally prioritizes nodes using $g(n) + h(n)$. Consider a variant that gives the heuristic twice as much weight and instead uses

$$f_2(n) = g(n) + 2h(n).$$

At each step, it removes a node with the smallest f_2 from the frontier. It returns a path when the goal is removed, not when the goal is first discovered. As in the other search questions, it reopens an expanded node if a lower-cost path to that node is found.

All edge costs are positive. Let C^* be the cost of an optimal path from the start to the goal. The heuristic is admissible and satisfies $h(G) = 0$. Write $g^*(n)$ for the optimal cost from the start to n and $h^*(n)$ for the optimal remaining cost from n to the goal.

You may use the following fact without proving it. Immediately before the algorithm removes the goal, some node n on an optimal path is still on the frontier with $g(n) = g^*(n)$.

- (a) Prove that $f_2(n) \leq 2C^*$ for the frontier node described above. Justify the inequalities in your argument.

Answer / work

Part 2: Programming practice

The starter code is in `midterm-practice/code/`. Complete the two learning modules, then run the provided experiment and public checks from the course repository root. The checks are available for practice, not as an additional PS2 requirement.

```
python midterm-practice/code/autograder.py
```

Task 1: Learning from experience

Connection to PS2. PS2 P2 evaluates a fixed policy from recorded transitions. PS2 P3 examines what Q-learning can learn when exploration does not reach an important reward. This task implements both settings on the same grid-MDP interface used in PS1.

In `passive_rl.py`, implement `direct_evaluation` and `td_evaluation`. The runner supplies completed episodes generated by a fixed policy on a small course-grid environment. Direct evaluation averages the discounted return from every visit to a state. TD evaluation processes the same transitions in chronological order, updating only the state being left. Initialize values to zero, keep terminal values at zero, and follow the supplied function contracts. Neither method changes the policy.

In `q_learning.py`, implement tabular Q-learning with ϵ -greedy exploration. For an observed transition (s, a, r, s') , use

$$\text{sample} = r + \gamma \max_{a'} Q(s', a'), \quad Q(s, a) \leftarrow (1 - \alpha)Q(s, a) + \alpha \text{sample}.$$

At a terminal transition, use $\text{sample} = r$. With probability ϵ , select a uniformly random action; otherwise select a greedy action, resolving ties in N, S, E, W order. Use only the supplied random generator, in the draw order specified in the starter. Record the training statistics using the provided `Stats` class.

Run `python midterm-practice/code/run_learning.py`. The first experiment compares direct and TD estimates from the same fixed-policy episodes. The second trains Q-learning on open, maze, and narrow using the settings in `constants.py`, then tests each greedy policy. The runner also repeats the maze experiment without exploration. The known model is used by the runner for evaluation only; neither learning implementation may inspect the transition model beyond the sampled transitions it receives or, for Q-learning, its calls to `world.step`.

Files to complete. Implement `passive_rl.py` and `q_learning.py`. Record the results below in your practice PDF.

PDF responses.

- (a) Include `practice_learning.png`, comparing the two passive estimates with the fixed policy's exact values. Explain in two sentences why finite-data estimates may differ even though the episodes are identical.

Answer / work

- (b) Include the Q-learning results table. Report goal-reaching performance and the greedy path length for each map, including the no-exploration maze run. Explain the latter result using PS2 P3, and distinguish evaluating a fixed policy from learning action values for policy improvement.

Answer / work